```
import pandas as pd

# Create Product Sales Dataset
data = {
    "Product": ["Laptop", "Phone", "Tablet", "Headphones", "Laptop",
                "Phone", "Tablet", "Headphones"],
    "Category": ["Electronics", "Electronics", "Electronics", "Accessories",
                 "Electronics", "Electronics", "Electronics", "Accessories"],
    "Price": [55000, 30000, None, 2500, 60000, None, 22000, 3000],
    "Quantity": [2, 5, 3, None, 1, 4, None, 6],
    "Rating": [4.5, 4.2, 4.0, None, 4.7, 4.1, 3.9, None]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df)


# ------------------------------------------------
# 1. DETECT MISSING VALUES
# ------------------------------------------------

print("\n1. Detect Missing Values:")
print(df.isnull())


# ------------------------------------------------
# 2. COUNT MISSING VALUES
# ------------------------------------------------

print("\n2. Count Missing Values:")
print(df.isnull().sum())


# ------------------------------------------------
# 3. FILLING MISSING VALUES
# ------------------------------------------------

df["Price"] = df["Price"].fillna(df["Price"].mean())
df["Quantity"] = df["Quantity"].fillna(df["Quantity"].mean())
df["Rating"] = df["Rating"].fillna(df["Rating"].mean())

print("\n3. Data After Filling Missing Values:")
print(df)


# ------------------------------------------------
# 4. GROUPING DATA
# ------------------------------------------------

print("\n4. Grouping Data by Category:")

grouped_data = df.groupby("Category")["Price"].mean()

print(grouped_data)
```

```
Original Data:
      Product     Category    Price  Quantity  Rating
0      Laptop  Electronics  55000.0       2.0     4.5
1       Phone  Electronics  30000.0       5.0     4.2
2      Tablet  Electronics      NaN       3.0     4.0
3  Headphones  Accessories   2500.0       NaN     NaN
4      Laptop  Electronics  60000.0       1.0     4.7
5       Phone  Electronics      NaN       4.0     4.1
6      Tablet  Electronics  22000.0       NaN     3.9
7  Headphones  Accessories   3000.0       6.0     NaN

1. Detect Missing Values:
   Product  Category  Price  Quantity  Rating
0    False     False  False     False   False
1    False     False  False     False   False
2    False     False   True     False   False
3    False     False  False      True    True
4    False     False  False     False   False
5    False     False   True     False   False
6    False     False  False      True   False
7    False     False  False     False    True

2. Count Missing Values:
Product     0
Category    0
```

```
Price       2
Quantity    2
Rating      2
dtype: int64

3. Data After Filling Missing Values:
      Product     Category    Price  Quantity    Rating
0      Laptop  Electronics  55000.0       2.0  4.500000
1       Phone  Electronics  30000.0       5.0  4.200000
2      Tablet  Electronics  28750.0       3.0  4.000000
3  Headphones  Accessories   2500.0       3.5  4.233333
4      Laptop  Electronics  60000.0       1.0  4.700000
5       Phone  Electronics  28750.0       4.0  4.100000
6      Tablet  Electronics  22000.0       3.5  3.900000
7  Headphones  Accessories   3000.0       6.0  4.233333

4. Grouping Data by Category:
Category
Accessories     2750.000000
Electronics    37416.666667
Name: Price, dtype: float64
```

Start coding or generate with AI.